

VP150 Recitation Class 8

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Elasticity

Elastic Modulus

$$F = k \Delta x$$

$$\text{elastic modulus} = \frac{\text{stress}}{\text{strain}}$$

pound force
 N/m^2 lbf/in^2
unitless *inches*

where stress is force per unit area, and strain is the fractional deformation due to the stress.

$$\text{Young's Modulus (tensile stress)} \quad Y = \frac{F_{\perp}/A}{\Delta l/l_0}$$

$$\text{Bulk Modulus (compressive stress)} \quad B = \frac{-\Delta p}{\Delta V/V_0}$$

$$\text{Shear Modulus (shear stress)} \quad S = \frac{F_{\parallel}/A}{x/h} = \frac{F_{\parallel}/A}{\sin \theta}$$

*Shear Modulus is usually much smaller.

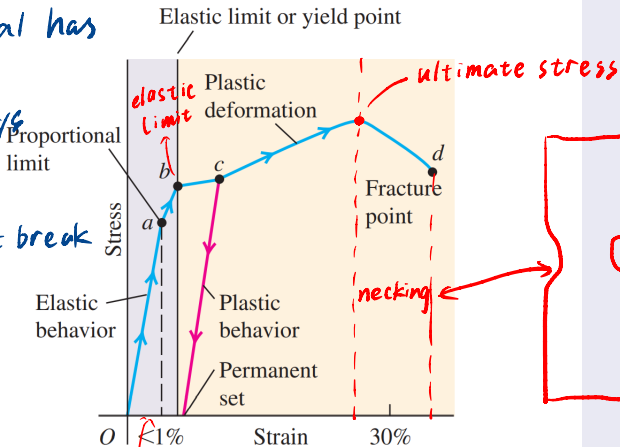


Elasticity and Plasticity

Plasticity

① material has a limit (not always linear behavior)

② It might break



similar to a spring with small k .

Basic Concepts

Density of Mass

$$\rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V}$$

→ around a particular point

Relative Density

$$r = \frac{\rho}{1000[\text{kg}/\text{m}^3]}$$

→ unitless
← water 4°C

Pressure

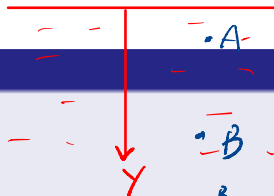
$$p = \frac{\Delta F_{\perp}}{\Delta A}$$

Pressure in Liquid

Equations

$$dp = -\rho g dy$$

$$\Delta p = \int_{p_1}^{p_2} dp$$



$$p_B - p_A = \int_A^B dp$$

$$= \int_A^B \rho(y) g dy$$

*The density of mass ρ and the gravity acceleration g might be function of depth y .

Pressure with Constant ρ and g

$$p = p_0 + \rho gh$$

atmospheric pressure: $1.01 \times 10^5 \approx 1 \times 10^5 [Pa]$

pressure in 1-meter-depth water: $9.8 \times 10^3 \approx 1 \times 10^4 [Pa]$

*U-shaped pipe



← surfaces w/ same height
(still)

Pascal's Law

Equations

$$\frac{F_2}{F_1} = \frac{A_2}{A_1} \quad \frac{F}{A} = P$$

*Verify it with

- (a) the definition of pressure,
- (b) work-kinetic energy theorem.

Archimedes' Principle

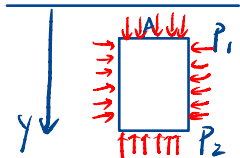
Buoyant Force

$$F_b = \rho_{\text{liquid}} V_{\text{immersed}} g$$

The buoyant force equals to the weight of the fluid displaced by the body in magnitude.

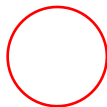
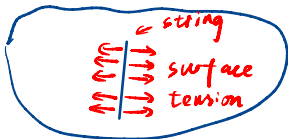
Practically, the density and the gravity acceleration can still be function of depth y . In this case, solve the buoyant force with integration.

*For objects with vertical side surfaces, buoyant force can be derived with $F_b = (p_2 - p_1)A$.



$$\Rightarrow F_b = (p_2 - p_1)A$$

Surface Tension



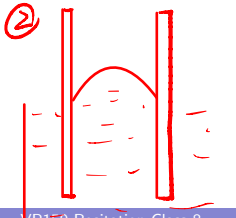
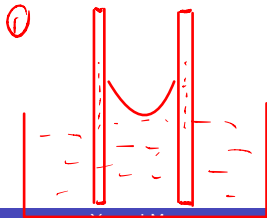
why does droplet always try to keep a sphere?

Origin

The attractive interaction between molecules of the liquid.

Meniscus and capillarity

- ① adhesion forces dominate – concave liquid surface
- ② cohesion forces dominate – convex liquid surface



Bernoulli's Equation

Requirements and Preparation Equations

Density of mass ρ is constant, which means the liquid cannot be compressed.

$$\delta W = p_1 dV - p_2 dV$$

$$dK = \frac{1}{2}\rho(v_2^2 - v_1^2)dV$$

$$dU_{grav} = \rho g(y_2 - y_1)dV$$

*Derive these equations yourself!

$$\delta W = dK + dU_{grav}$$

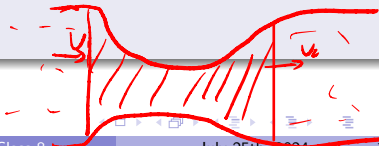
$$p_1 dV - p_2 dV = \frac{1}{2}\rho(v_2^2 - v_1^2)dV + \rho g(y_2 - y_1)dV$$

$$\xrightarrow{/dV} p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

Equations

$$p + \frac{1}{2}\rho v^2 + \rho g y = \text{const} \quad \vec{v} \perp \text{surface}$$

*Verify it with work-kinetic energy.



Viscosity and Turbulence

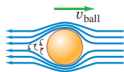
Definition

Viscosity is internal friction in a fluid.

Turbulence is irregular chaotic flow that is no longer laminar.

Application

(a) Motion of air relative to a nonspinning ball



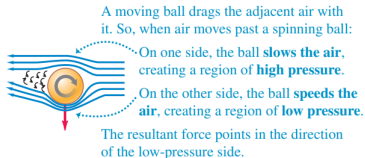
(b) Motion of a spinning ball

This side of the ball moves opposite to the airflow.



This side moves in the direction of the airflow.

(c) Force generated when a spinning ball moves through air



Newton's Law of Gravitation

Definition

Force of gravitational attraction

$$\vec{F}_{12} = -G \frac{m_1 m_2}{r_{12}^2} \frac{\vec{r}_{12}}{|\vec{r}_{12}|}$$

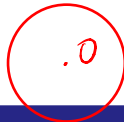
Gravitational Field

$$\vec{E}_G = -G \frac{M}{r^2} \frac{\vec{r}}{|\vec{r}|}$$



$\vec{E}_G$ is a property of M .

Newton's Law of Gravitation



← treat it as a mass point at O to calculate $\vec{F}_G$ outside the uniform sphere.

Properties

Newton's Third Law

$$\vec{F}_{12} = -\vec{F}_{21}$$

The Superposition Principle

Properties

Gravitation from Uniform Spheres

For a spherically symmetric distribution of mass in a region Ω , the gravitational field outside of Ω is as if the whole mass was concentrated at the center of Ω .

Gravitation inside Uniform Spherical Shells

The gravitation field inside a uniform spherical shell holds the magnitude of 0.

Gravitational Potential Energy

Definition

Gravitational Potential Energy

Usually, choose $r \rightarrow \infty$ as zero potential point

$$U(r) = -G \frac{Mm}{r} + C$$

$$\Rightarrow U(r) = -G \frac{Mm}{r}$$

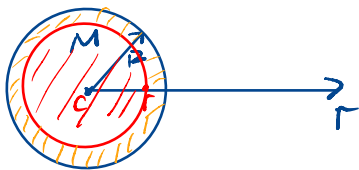
The Zero Potential Point can be selected arbitrarily. In another word, C can be set (probably by MK) arbitrarily.

* Derive the equation yourself with integration on $\vec{F}_{12}$!

$r \rightarrow \infty$ as 0

$$\begin{aligned} U(r) &= - \int_{\infty}^r F(r) dr = - \int_{\infty}^r - \frac{GMm}{r^2} dr = \int_{\infty}^r GMm r^{-2} dr \\ &= - \frac{GMm}{r} \Big|_{\infty}^r \end{aligned}$$

Gravitational Potential Energy



$$F = \begin{cases} ? & 0 \leq r \leq R \\ ? & r > R \end{cases}$$

Example

Force Graph (versus distance)

Draw the ~~Gravitational Potential Energy Graph (versus distance)~~ of a uniform ^{solid} ~~spherical shell~~ ^{ball} with radius R and Mass M . Take an infinitely far point as the zero potential point.

For $0 \leq r \leq R$, only red part of the ball has contribution to the $\vec{F}_G$ at r .

$$F = \frac{GM_r m}{r^2} \quad M_r = \frac{M}{R^3} \cdot r^3 \Rightarrow F = \frac{GMm}{R^3} \cdot r \propto r$$

$$R < r \Rightarrow F = \frac{GMm}{r^2} \propto r^{-2}$$

Gravitational Potential Energy

$$V(r) = \begin{cases} ? & 0 \leq r \leq R \\ ? & r > R \end{cases}$$

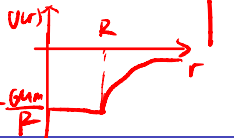
$$F(r) = \begin{cases} -\frac{GMm}{R^2} & 0 \leq r \leq R \\ -\frac{GMm}{r^2} & r > R \end{cases}$$

$$F(r) = \begin{cases} 0 & 0 \leq r \leq R \\ -\frac{GMm}{r^2} & r > R \end{cases}$$

Example

Draw the Gravitational Potential Energy Graph (versus distance) of a uniform spherical shell with radius R and Mass M . Take an infinitely far point as the zero potential point.

② solid uniform spherical ball

$$V(r) = \begin{cases} -\frac{GMm}{r} & r > R \\ -\frac{GMm}{R} & 0 \leq r \leq R \end{cases}$$


$$V(r) = \begin{cases} -\frac{GMm}{r} & r > R \\ -\frac{GMm}{R} + \frac{GMm}{2R^3}(r^2 - R^2) & 0 \leq r \leq R \end{cases}$$


Gravitation

Example

(12 points) Consider a solid sphere of water sitting in deep space. The sphere has a uniform density of $\rho = 1000 \text{ kg/m}^3$ and a radius of $R = 100 \text{ km}$.

Using what we know about gravity, find a formula for the force of gravity everywhere within the volume of the sphere. Find the pressure of the water at a distance $r = 1 \text{ meter}$ from the center of the sphere.

Escape Speed

$$U(R) = -\frac{GMm}{R}$$

infinity
0



Definition

initial
energy

$$\frac{1}{2}mv_{\text{esc}}^2 - G\frac{Mm}{R} = 0$$

energy at infinity

conservation of
energy

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

$$\frac{1}{2}mV_{\text{inf}}^2 - \frac{GMm}{R_{\text{inf}}} = 0$$

$$V_{\text{inf}} = 0$$

$$R_{\text{inf}} \rightarrow \infty$$

Properties

What may the trajectory be if speed larger than escape speed?

Satellite Motion

Definition

$$G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

→ circular motion

$$v = \sqrt{\frac{GM}{r}}$$

ellipse
oftentimes

$$T = \frac{2\pi r}{v} = 2\pi \frac{r^{3/2}}{\sqrt{GM}}$$

$r \rightarrow$ semimajor axis

半长轴

applies for both circle and ellipse

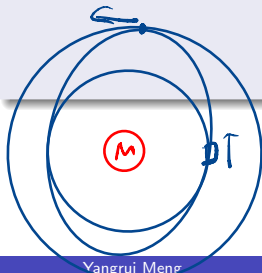
Kepler's Laws

Definition

Each planet moves in an elliptical orbit with the Sun at one of the focal points.

The line from the Sun to a given planet sweeps out equal areas in equal times. (Conservation of Angular Momentum)

The period of motion of a planet is proportional to the 3/2-power of the major axis length of its orbit



$$T = 2\pi \frac{a^{3/2}}{\sqrt{GM}}$$

Reference

Zhemin Qu, 2023SU VP150 RC.

Mateusz Krzyzosiak, 2024SU VP150 Lec.